

Auburn-Regions Data Hackathon

Participant Guide and Data Dictionary

Auburn University | Sponsored by Regions Bank

Challenge brief

Build a reproducible decision-support solution that estimates the probability of material payment difficulty for each applicant in the hidden evaluation set, then translate those estimates into a responsible lending recommendation. The data are simulated and do not represent Regions Bank customers, underwriting policy, or financial assumptions.

- Predict a probability between 0 and 1 for every evaluation CustomerID.
- Show how relational history improves, or does not improve, the applicant-level model.
- Explain business trade-offs, model limitations, and responsible-use checks.
- Provide code and environment details that allow organizers to rerun the submission.

Data release

Split	Applicants	Outcome included
Training	30,000	Yes
Validation	7,500	Yes
Evaluation	7,500	No

The generated default rate is approximately 12.1%. Applicant files share CustomerID with `credit_account_history.csv`, `prior_lending_activity.csv`, and `responsible_use_audit.csv`.

- `application_train.csv` - primary labeled development data.
- `application_validation.csv` - labeled holdout for local checking.
- `application_evaluation.csv` - unlabeled competition records.
- `credit_account_history.csv` - up to six historical account records per applicant.
- `prior_lending_activity.csv` - up to four prior requests per applicant.
- `responsible_use_audit.csv` - audit-only groups. Do not use them as model inputs or lending criteria.

Required submission

1. Create `predictions.csv` with exactly two columns: CustomerID and PredictedDefaultProbability.
2. Include every evaluation CustomerID exactly once; do not include training or validation IDs.
3. Use decimal probabilities in [0, 1]. Hard labels are not accepted.
4. Submit code, dependency information, a short executive summary, and presentation slides alongside the prediction file.

Prediction performance is evaluated using ROC AUC and contributes to the weighted overall score. The same overall score also incorporates business relevance, engineering and reproducibility, communication, and responsible analytics. A technically valid prediction file is necessary but not sufficient to place.

Awards and overall ranking

All teams compete for one overall ranking. Judges score each of the five dimensions on a 0-100 scale; the published weights produce a single overall score out of 100. The three highest-ranked eligible teams receive the following awards.

Overall rank	Award	Details
1st Place	\$12,000	\$3,000 per student for four team members
2nd Place	\$6,000	\$1,500 per student for four team members
3rd Place	Bicycles	Bicycle award; fulfillment details communicated by the organizers

Judging dimension	Weight	Evidence
Predictive analytics	25%	Hidden-evaluation AUC, validation discipline, and calibration discussion
Business decision strategy	25%	Actionable recommendation and risk-growth trade-offs
Data engineering and reproducibility	20%	Relational features, clean pipeline, and rerun success
Storytelling and communication	15%	Clear evidence, visuals, and presentation
Responsible analytics	15%	Appropriate audit use, limitations, and monitoring plan

Overall score = 25% Predictive Analytics + 25% Business Decision Strategy + 20% Data Engineering and Reproducibility + 15% Storytelling and Communication + 15% Responsible Analytics. Ties are reviewed first on Responsible Analytics, then Data Engineering and Reproducibility, then Predictive Analytics.

Competition integrity and responsible use

- External labeled data, source-record lookup, recovered public labels, or manual label assignment are prohibited.
- AuditGenderGroup and AuditAgeBand are supplied only for group-level evaluation and must not enter a model or approval rule.
- Teams must disclose all external libraries, pretrained models, and non-release data used.
- Finalists must provide runnable code; organizers may rerun work from the released files in a clean environment.
- The task is educational. Do not interpret results as a deployable underwriting model.

Starter workflow

- Run starter/python_starter.py from the package root for a tested numeric baseline.
- Use submission/prediction_template.csv as the authoritative evaluation-ID roster.
- Install dependencies from starter/requirements.txt or starter/environment.yml.
- Verify the ZIP and individual files using SHA256SUMS.txt before beginning work.

Data dictionary

File	Variable	Type / units	Description
application_*	CustomerID	text	Synthetic join key shared across tables.
application_*	ApplicationMonth	YYYY-MM	Month of the simulated application.
application_*	ProductType	category	Installment, revolving, or other credit product.
application_*	LoanPurpose	category	Simulated primary use of requested credit.
application_*	ApplicantAge	years	Age at application.
application_*	EmploymentYears	years	Approximate continuous employment tenure; may be missing.
application_*	AnnualIncome	USD-like units	Simulated annual household income; values are not actual dollars from a bank.
application_*	RequestedAmount	USD-like units	Requested credit amount.
application_*	AnnualPayment	USD-like units	Estimated annual scheduled payment.
application_*	GoodsValue	USD-like units	Estimated value associated with financed goods, where applicable.
application_*	Children	count	Number of dependent children, capped at six.
application_*	HouseholdSize	count	Number of people in the household.
application_*	EducationLevel	category	Recoded education category.
application_*	MaritalStatus	category	Recoded marital or partnership status.
application_*	HousingStatus	category	Current housing arrangement.
application_*	IncomeSource	category	Primary income-source group.
application_*	OccupationGroup	category	Broad occupational group; may be missing.

File	Variable	Type / units	Description
application_*	OwnsVehicle	category	Whether the applicant owns a vehicle.
application_*	OwnsHome	category	Whether the applicant reports real-property ownership.
application_*	MarketTier	category	Synthetic market-size and volatility grouping.
application_*	ResidenceWorkMismatch	0/1	Indicator that residence and work market differ.
application_*	VehicleAgeYears	years	Age of owned vehicle; missing can also mean unavailable.
application_*	VendorScoreAtlas	300-850	Simulated third-party risk score; higher indicates lower modeled risk.
application_*	VendorScoreBeacon	300-850	Second simulated third-party risk score.
application_*	VendorScoreCedar	300-850	Third simulated third-party risk score.
application_*	CreditInquiries12M	count	Recent credit inquiries.
application_*	PhoneTenureYears	years	Approximate tenure of current phone contact.
application_*	CommunityLatePaymentRate	0-1	Noisy contextual late-payment indicator; not an individual payment record.
application_*	LoanToIncomeRatio	ratio	RequestedAmount / AnnualIncome.
application_*	PaymentToIncomeRatio	ratio	AnnualPayment / AnnualIncome.
application_*	IncomePerHouseholdMember	USD-like units	AnnualIncome / HouseholdSize.
application_*	DataCompletenessScore	0-1	Share of applicant fields present before final controlled masking.
application_*	PriorAccountCount	count	Count of supplied credit-account history records.
application_*	ActiveAccountCount	count	Count of supplied accounts currently open.
application_*	OutstandingBalance	USD-like units	Sum of current balances in supplied account history.
application_*	OverdueAccountCount	count	Accounts with past-due days or overdue balance.
application_*	CreditHistoryYears	years	Age of oldest supplied account record.
application_*	PriorApplicationCount	count	Count of supplied prior lending requests.
application_*	PriorApprovalRate	0-1	Share of supplied prior requests approved.
application_*	AveragePriorRequestedAmount	USD-like units	Mean requested amount in prior requests.
application_*	DebtToIncomeRatio	ratio	(OutstandingBalance + RequestedAmount) / AnnualIncome.
train/validation	DefaultFlag	0/1	Newly generated outcome. 1 denotes material payment difficulty in the simulated performance window.
credit_account_history	AccountRecordID	text	Synthetic account-record key.
credit_account_history	AccountStatus	category	Open, closed, transferred, charged off, or other.
credit_account_history	AccountType	category	Broad account product.
credit_account_history	OpenedMonthsAgo	months	Months between account opening and current application.
credit_account_history	ScheduledEndMonthOffset	months	Scheduled end relative to application; positive is after application.
credit_account_history	OriginalBalance	USD-like units	Original account balance or limit.
credit_account_history	CurrentBalance	USD-like units	Outstanding balance at application.
credit_account_history	OverdueBalance	USD-like units	Past-due balance.
credit_account_history	DaysPastDue	days	Days past due at the reference date.
credit_account_history	TimesExtended	count	Number of reported extensions or prolongations.
prior_lending_activity	PriorRequestID	text	Synthetic prior-request key.
prior_lending_activity	DecisionStatus	category	Approved, declined, withdrawn, offer not used, or other.
prior_lending_activity	ProductCategory	category	Broad prior product.
prior_lending_activity	RequestedAmount	USD-like units	Amount requested in prior request.
prior_lending_activity	ApprovedAmount	USD-like units	Amount approved or offered.
prior_lending_activity	ScheduledPayment	USD-like units	Scheduled payment when available.
prior_lending_activity	TermMonths	months	Requested or approved term.
prior_lending_activity	DecisionMonthsAgo	months	Months between prior decision and current application.
prior_lending_activity	ChannelGroup	category	Broad acquisition or servicing channel.
prior_lending_activity	PricingBand	category	Synthetic relative pricing band.
prior_lending_activity	DownPaymentRate	0-1	Down payment divided by associated purchase amount, where available.
responsible_use_audit	AuditGenderGroup	category	Audit-only simulated group; prohibited as a model or approval input.
responsible_use_audit	AuditAgeBand	category	Audit-only age band for performance comparisons.